

Assignment - 1

Analysis of One Published Biological Dataset Using the Data Science Workflow

Human Microbiome Project (HMP) 16S rRNA Dataset

This assignment analyzes a published biological dataset from the Human Microbiome Project (HMP) using the data science workflow. The selected dataset is the HMP healthy-human 16S rRNA community profiling dataset. It is substantially different from a transcriptomics dataset because it describes microbial community composition rather than host gene expression. The dataset is therefore suitable for demonstrating how data science can be applied to microbiome data.

The HMP characterized microbial communities from healthy adults across multiple body habitats. The published HMP framework reported 5,177 microbial taxonomic profiles from 16S rRNA genes from 242 healthy adults sampled at 15 or 18 body sites, together with more than 3.5 terabases of metagenomic sequence. A related HMP study examined 4,788 specimens from 242 screened and phenotyped adults and found strong body-site-specific microbial specialization.

1. Research Question

How does the composition and diversity of the human microbiome differ among different body sites in healthy adults?

Specific questions include:

- Which bacterial taxa are most abundant at different human body sites?
- Do microbial communities from the same body site show similar composition?
- Are some body sites characterized by higher microbial diversity than others?
- Can multivariate analysis distinguish samples according to their body site?

2. Data Source

The data source is the Human Microbiome Project (HMP), an NIH-supported project that generated standardized microbial community profiles and metagenomic resources from healthy human participants.

For this analysis, the selected data type is the HMP 16S rRNA gene community-profiling dataset. The HMP framework paper reports 5,177 microbial taxonomic profiles generated from 16S rRNA genes from 242 healthy adults sampled at 15 or 18 body sites. The HMP Data Analysis and Coordination Center (DACC) served as the central repository for the project's 16S, whole-metagenome, and reference genome sequence information.

Primary publication: The Human Microbiome Project Consortium. (2012). A framework for human microbiome research. *Nature*, 486, 215–221. doi:10.1038/nature11209.

Related biological findings publication: The Human Microbiome Project Consortium. (2012). Structure, function and diversity of the healthy human microbiome. *Nature*, 486, 207–214. doi:10.1038/nature11234.

3. Data Type

The selected dataset is 16S rRNA gene amplicon sequencing data. The 16S rRNA gene contains conserved and variable regions that can be used to characterize bacterial community composition and assign sequences to taxonomic groups.

The resulting biological data can be represented as a feature-by-sample abundance matrix, where rows represent microbial taxa or operational taxonomic units (OTUs), columns represent samples, and values represent the abundance or relative abundance of each microbial feature.

The dataset also contains sample metadata such as body-site information and participant-related variables, allowing microbial composition to be compared across anatomical habitats.

4. Data Collection Method

Healthy adult participants were carefully screened and phenotyped before sampling. Depending on sex, participants were sampled at 15 or 18 body sites, with samples collected up to three times using a common sampling protocol.

The general experimental workflow was:

Human body-site sample → microbial DNA extraction → 16S rRNA gene amplification → high-throughput sequencing → sequence quality control → taxonomic profiling → microbial abundance matrix

The HMP 16S community-profiling protocols were evaluated across sequencing centers using platforms including ABI 3730 and 454 FLX Titanium sequencing.

5. Data Cleaning Steps

Before microbiome analysis, sequencing reads and sample metadata need to be quality controlled and standardized. The following data-cleaning steps would be applied:

1. Check sample identifiers and body-site labels for consistency.
2. Remove low-quality sequencing reads using quality-score thresholds.
3. Remove primer and adapter sequences when present.
4. Remove or flag chimeric sequences that can arise during PCR amplification.
5. Filter features with extremely low abundance or very low prevalence when appropriate.
6. Assign sequences to microbial taxa using a suitable reference database and classification method.
7. Check for contaminant sequences and technical artifacts.
8. Identify missing or inconsistent metadata and resolve them before statistical comparison.
9. Convert appropriate abundance measurements to relative abundance or use a suitable normalization method for the planned analysis.
10. Inspect sample-level quality and identify potential outliers before diversity and ordination analyses.

6. Exploratory Analysis

Exploratory Data Analysis (EDA) would be used to understand the microbial community structure before formal statistical testing.

- Taxonomic abundance summaries: calculate the relative abundance of major bacterial phyla, genera, or other taxonomic levels.
- Alpha diversity: measure within-sample diversity using indices such as Shannon or Simpson diversity.
- Beta diversity: calculate between-sample dissimilarity using a measure such as Bray–Curtis dissimilarity.
- Sample relationships: examine whether samples from the same body site are more similar to each other than samples from different body sites.
- Ordination: use Principal Coordinates Analysis (PCoA) to visualize major patterns in community composition.
- Clustering: use hierarchical clustering to identify groups of samples with similar microbial community profiles.

The published HMP analysis found strong niche specialization among body habitats. Oral and stool communities were particularly diverse, whereas vaginal communities were relatively simple. This provides a biologically meaningful expectation for exploratory analysis of the dataset.

7. Analysis or Modeling Method

The primary analysis would use alpha diversity, beta diversity, and ordination. These methods are appropriate because the goal is to characterize microbial community structure and determine whether body site is associated with microbiome composition.

7.1 Alpha Diversity

Alpha-diversity indices quantify diversity within individual samples. Shannon diversity incorporates both richness and evenness, while Simpson-based measures emphasize the distribution of abundant taxa.

7.2 Beta Diversity

Beta diversity measures differences in microbial composition between samples. Bray–Curtis dissimilarity can be used to quantify differences in taxon abundance between samples.

7.3 Principal Coordinates Analysis

PCoA can transform the beta-diversity distance matrix into a low-dimensional representation. Samples that have similar microbial communities should appear close together, while highly different communities should appear farther apart.

7.4 Statistical Comparison

If formal hypothesis testing is performed, microbial community differences between body sites can be tested using an appropriate multivariate method such as PERMANOVA, with body site as the explanatory factor. The analysis should account for the compositional nature of microbiome

abundance data and should be interpreted together with effect sizes and appropriate multiple-testing procedures where relevant.

8. Visualization Approach

The following visualizations would communicate the findings effectively:

- Stacked bar charts showing relative abundance of major microbial taxa across body sites.
- Boxplots showing alpha-diversity values for different body sites.
- PCoA scatter plots showing beta-diversity relationships among samples.
- Heatmaps showing the abundance of selected taxa across representative samples.
- Hierarchical clustering dendrograms showing relationships among microbial communities.
- Rank-abundance plots showing how dominant and less abundant taxa contribute to community structure.

9. Biological Interpretation

The expected biological interpretation is that microbial communities are strongly shaped by the anatomical and ecological environment of the body site. Different habitats provide different physical and chemical conditions, which favor different microbial communities.

- Oral and stool communities are expected to contain high microbial diversity and many different taxa.
- Vaginal communities are expected to be comparatively less diverse and dominated by a smaller set of characteristic organisms.
- Skin communities are expected to vary substantially between individuals and body locations.
- Samples from the same body habitat are expected to be more similar in microbial composition than samples from unrelated body habitats.

These patterns would indicate strong niche specialization in the healthy human microbiome. The published HMP study reported that microbial diversity and abundance vary substantially among body habitats and that microbial community structure can be strongly habitat-specific. It also showed that functional pathway profiles can be more stable among individuals than taxonomic composition.

10. Limitations and Challenges

16S resolution: 16S rRNA sequencing generally provides lower taxonomic resolution than whole-metagenome sequencing and may not reliably distinguish closely related species or strains.

Relative abundance: Most microbiome abundance data are compositional, meaning that an increase in one taxon can change the relative proportions of others even when their absolute abundances do not change.

PCR and sequencing bias: Primer choice, amplification efficiency, sequencing technology, and other technical factors can influence observed community composition.

Sampling variation: Microbial communities can vary across body locations, individuals, and time, so a single sample may not fully represent an individual's microbiome.

Host and environmental factors: Diet, age, sex, geography, medication, lifestyle, and other factors may influence microbiome composition.

Contamination: Low-biomass samples can be especially vulnerable to contamination, making careful quality control important.

Unequal sample representation: Different body sites may have different numbers of usable samples, affecting statistical power and comparisons.

Association versus causation: An association between a microbial taxon and a body site does not demonstrate that the microorganism causes a host phenotype.

Generalizability: The original HMP cohort was focused on healthy adults in the United States, so findings may not represent all populations or disease states.

Overall Data Science Workflow

11. Define the biological question about microbiome differences among human body sites.
12. Identify the HMP 16S rRNA dataset and associated sample metadata.
13. Obtain and organize the sequencing-derived taxonomic abundance data.
14. Perform sequencing quality control, feature filtering, taxonomic assignment, and metadata cleaning.
15. Explore taxonomic composition, alpha diversity, beta diversity, correlations, and sample structure.
16. Apply beta-diversity analysis, PCoA, clustering, and appropriate statistical testing.
17. Evaluate whether observed community differences are consistent with body-site biology.
18. Visualize findings using abundance plots, diversity plots, heatmaps, PCoA, and clustering.
19. Interpret microbial patterns in terms of ecological niche specialization and human biology.
20. Report conclusions, limitations, and implications.

Conclusion

The Human Microbiome Project 16S rRNA dataset is an appropriate published biological dataset for demonstrating the data science workflow in bioinformatics. Unlike a gene-expression dataset, it focuses on the composition and diversity of microbial communities associated with different human body sites.

Through data cleaning, exploratory analysis, alpha- and beta-diversity analysis, PCoA, clustering, visualization, and biological interpretation, the dataset can be transformed into meaningful evidence about the organization of the healthy human microbiome. The HMP provides a strong example of how large-scale biological data can be used to discover reproducible patterns of biological and ecological variation.

References

- The Human Microbiome Project Consortium. (2012). A framework for human microbiome research. *Nature*, 486, 215–221. <https://doi.org/10.1038/nature11209>
- The Human Microbiome Project Consortium. (2012). Structure, function and diversity of the healthy human microbiome. *Nature*, 486, 207–214. <https://doi.org/10.1038/nature11234>

- Jumpstart Consortium Human Microbiome Project Data Generation Working Group. (2012). Evaluation of 16S rDNA-based community profiling for human microbiome research. PLOS ONE, 7(6), e39315. <https://doi.org/10.1371/journal.pone.0039315>
- Human Microbiome Project Data Analysis and Coordination Center (HMP DACC). Human Microbiome Project data resources.
- Introduction to Data Science, Module 1: Data Science and Cloud Computing in Bioinformatics. Course lecture material.